

PHL 1010 Tutorial I
18/08/2025 (Monday)

- Consider an ordered triplet $(\sqrt{a_y a_z}, \sqrt{a_z a_x}, \sqrt{a_x a_y})$ constructed from a vector $\vec{a} \equiv (a_x, a_y, a_z)$. Find it out whether the ordered triplet $(\sqrt{a_y a_z}, \sqrt{a_z a_x}, \sqrt{a_x a_y})$ represents a vector? For this, you may choose as an example, $\vec{a} \equiv (1/2, \sqrt{3}/2, 0)$ and the rotation of coordinate system about z-axis by 45° .
- In a Cartesian coordinate (x, y, z) system, a vector $\vec{V}$ has components given by (V_x, V_y, V_z) . This coordinate system is now rotated clockwise by 240° about an axis passing through the origin and the point $(2, 2, 2)$. What are the components of the vector $\vec{V}$ in the new coordinate system?
- Find out the components of the unit vector $\hat{n}$, normal to the shaded plane as shown in the figure.
- In a given region, a scalar field is defined by $\phi = (x^2 y z + 4 x z^2)$. If a person, standing at a point $(1, -2, -1)$, moves in the direction defined by $2\hat{x} - \hat{y} - 2\hat{z}$, what would be the differential change in the field as observed by him?
 - Evaluate $\vec{\nabla} f(r)$, where r has usual meaning. Use the result to evaluate $\vec{\nabla}(1/r)$.
 - Evaluate $\vec{\nabla} \times (\vec{r} f(r))$.
- A vector field is defined as $\vec{F} = x^2 \hat{x} - xz \hat{y} - y^2 \hat{z}$. Calculate the circulation of $\vec{F}$ along the closed path shown in the fig (2).
- Griffith Problem: 1.20 {prove the identities (iv) and (vi) only}/pp 22; Problem: 1.29/pp 28.

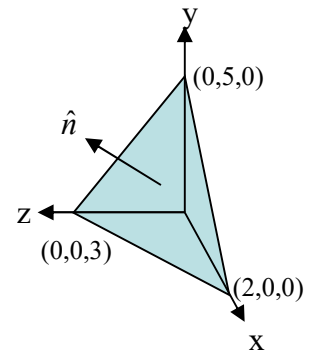


Figure 1

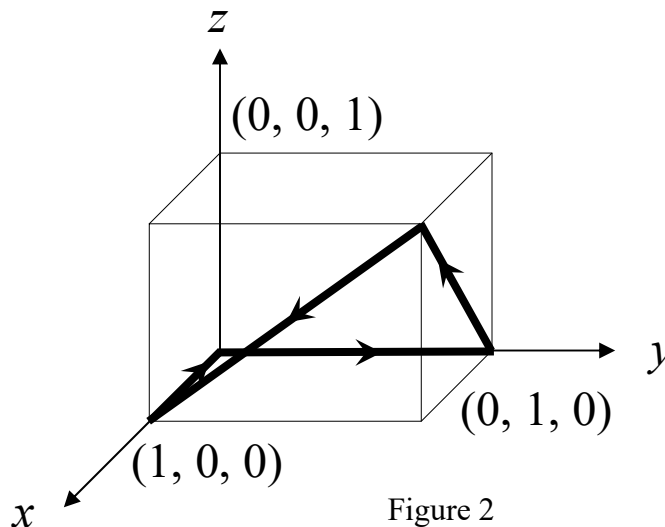


Figure 2